

CS3807 – Deep Learning Laboratory

Experiment 7: Autoencoders on MNIST

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1 Objective

The goal of this experiment was to build and compare four autoencoder variants on the MNIST dataset: a fully connected autoencoder, a convolutional autoencoder, a denoising convolutional autoencoder, and a variational autoencoder. Each model was trained to reconstruct handwritten digits, and their performance was evaluated using MSE, MAE, and SSIM. The VAE was also used to generate new digits and to explore the latent space.

2 Dataset and Experimental Setup

MNIST handwritten digits (28×28 grayscale) were used. Pixel values were scaled to $[0, 1]$. A subset of 10,000 training images and 2,000 test images was used for all experiments. For the fully connected model, images were flattened to 784-dimensional vectors. Convolutional models kept the $28 \times 28 \times 1$ shape. All models were trained with the Adam optimizer (learning rate 10^{-3}) and batch size 128.

3 Fully Connected Autoencoder

3.1 Architecture

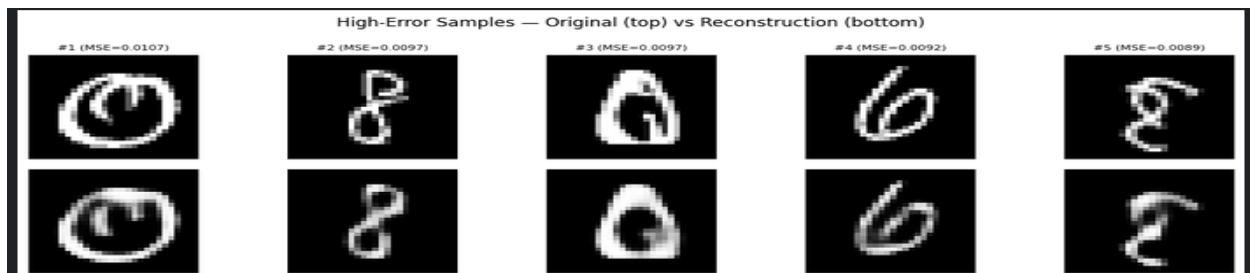
The encoder had dense layers of 128 and 32 units (ReLU), followed by a 16-dimensional latent layer. The decoder mirrored this: dense 32, dense 128, and a 784-unit output layer with sigmoid activation. The loss was binary cross-entropy.

3.2 Results

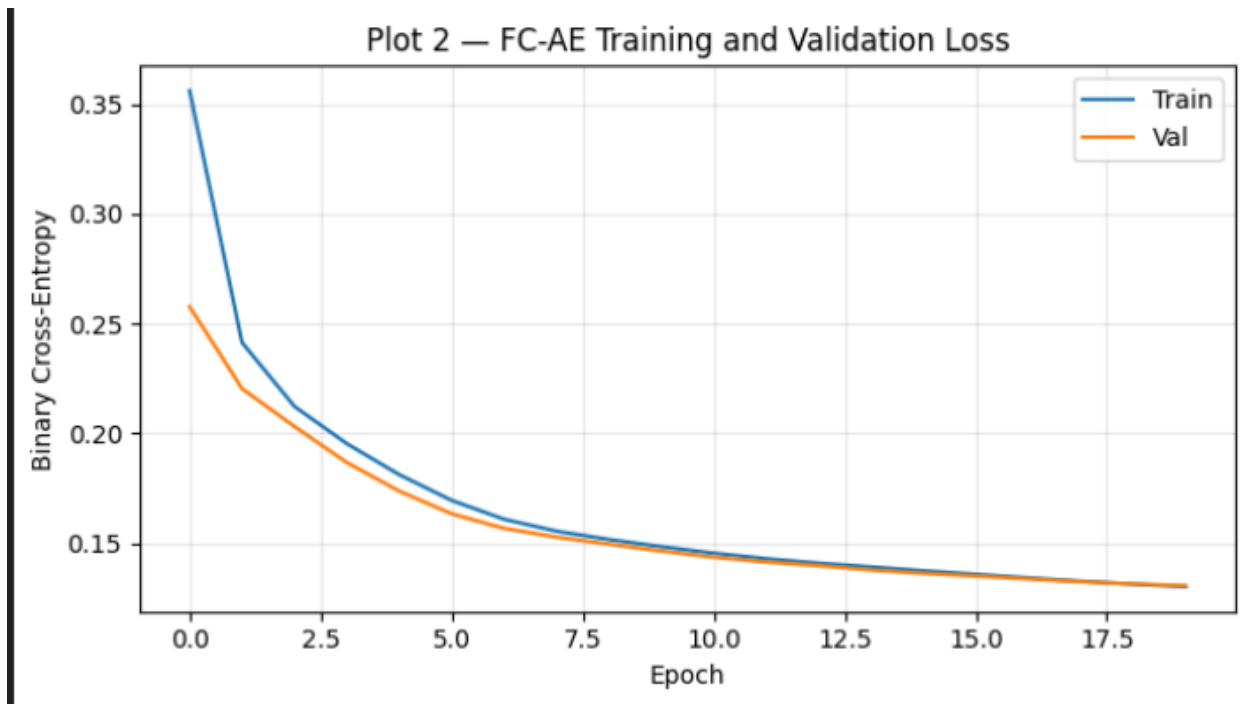
Training for 20 epochs took 9.7 seconds. Test metrics are shown in Table 1.

Fully connected autoencoder test results.

Metric	Value
MSE	0.02253
MAE	0.05921
SSIM	0.7331
Parameters	211,040
Training time	9.7 s



Original (top) and reconstructed (bottom) test digits from the fully connected autoencoder.



Training and validation loss for the fully connected autoencoder.

3.3 Inference

The model converges smoothly. Training and validation loss track each other closely, so there is no overfitting. Reconstructions preserve the overall digit shape but lose fine details. Thin strokes often appear blurred, and some digits (like 8 or 9) lose their inner loops. The latent dimension of 16 is small, so the model has to prioritise coarse structure over high-frequency detail.

4 Convolutional Autoencoder

4.1 Architecture

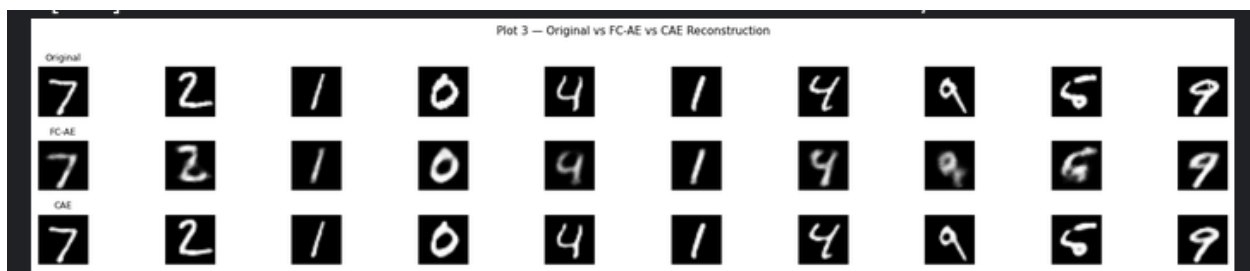
The encoder used two convolutional blocks: Conv2D(32,3) + MaxPool, then Conv2D(64,3) + MaxPool, followed by a Conv2D(64,3) latent feature map. The decoder used UpSampling + Conv2D(32,3) and UpSampling + Conv2D(1,3,sigmoid). All convolutions used 'same' padding and ReLU activations (except the output).

4.2 Results

Training took 16.6 seconds. Test metrics are in Table 2.

Convolutional autoencoder test results.

Metric	Value
MSE	0.00251
MAE	0.01491
SSIM	0.9735
Parameters	74,497
Training time	16.6 s



Original images (left), FC-AE reconstructions (middle), and CAE reconstructions (right).

4.3 Comparison with FC-AE

The CAE reduces MSE by a factor of about 9 and improves SSIM from 0.733 to 0.974. It also uses roughly one third of the parameters. Convolutions preserve spatial locality, so the decoder can rebuild edges and strokes more accurately. The FC-AE flattens the image and loses neighbour relationships, which shows up as blur.

5 Denoising Autoencoder

5.1 Noise Types

Two corruption types were used: Gaussian noise with $\sigma \in \{0.1, 0.2, 0.3\}$, and salt-and-pepper noise with $p \in \{0.05, 0.10, 0.20\}$. The same CAE architecture was trained with noisy inputs and clean targets.

5.2 Results

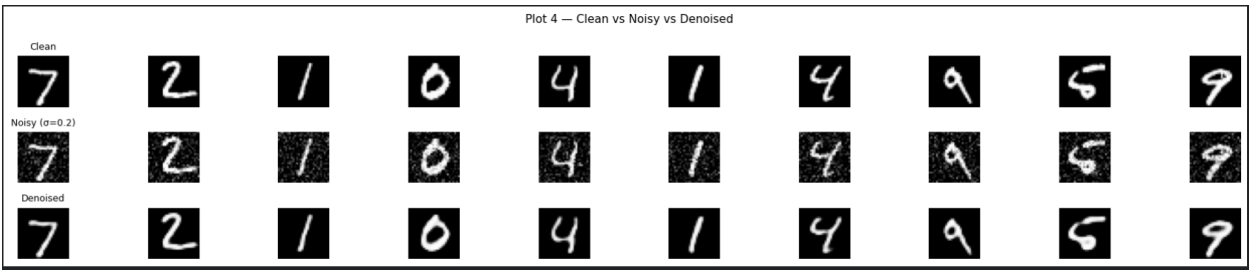
Denoising performance for Gaussian noise is in Table 3. Salt-and-pepper results are in Table 4.

Denoising CAE performance on Gaussian noise.

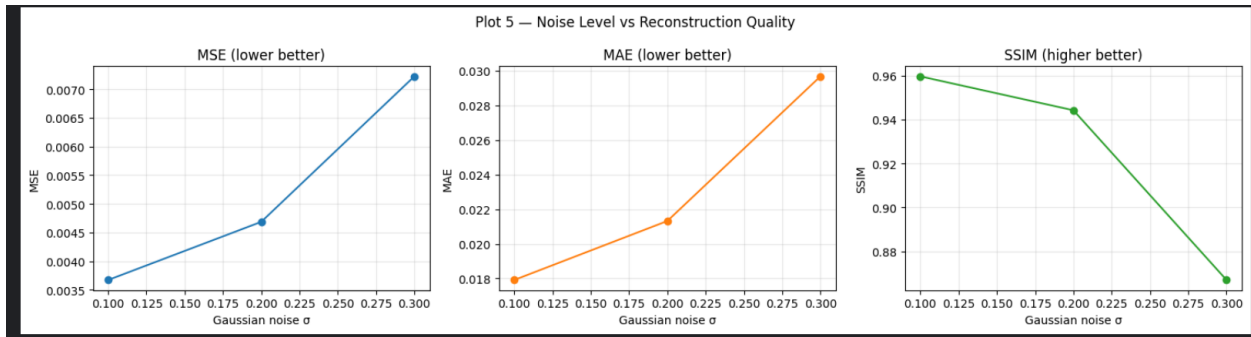
σ	MSE	MAE	SSIM
0.1	0.0037	0.0179	0.9596
0.2	0.0047	0.0213	0.9442
0.3	0.0072	0.0297	0.8670

Denoising CAE performance on salt-and-pepper noise.

p	MSE	MAE	SSIM
0.05	0.0049	0.0214	0.9280
0.10	0.0069	0.0272	0.8735
0.20	0.0133	0.0435	0.7372



Clean (left), noisy with $\sigma=0.2$ (middle), and denoised (right) digits



Noise level vs MSE, MAE, and SSIM for Gaussian noise.

5.3 Inference

Increasing corruption degrades reconstruction quality. At $\sigma=0.3$ the SSIM drops to 0.867, but the model still recovers the overall digit shape. The denoiser was trained only on Gaussian noise, yet it also reduces salt-and-pepper noise reasonably well at low levels. This suggests it learned a general smoothing prior rather than memorising one specific corruption pattern.

6 Variational Autoencoder

6.1 Architecture

The encoder had two convolutional layers (stride 2) followed by a dense layer that output μ and $\log \sigma^2$ (both 2-dimensional). The decoder mapped a 2D latent vector through a dense layer and two transposed convolutions to a $28 \times 28 \times 1$ output.

6.2 Loss Function

The VAE loss combines reconstruction loss (binary cross-entropy) and KL divergence:

$$L_{VAE} = L_{rec} + L_{KL}$$

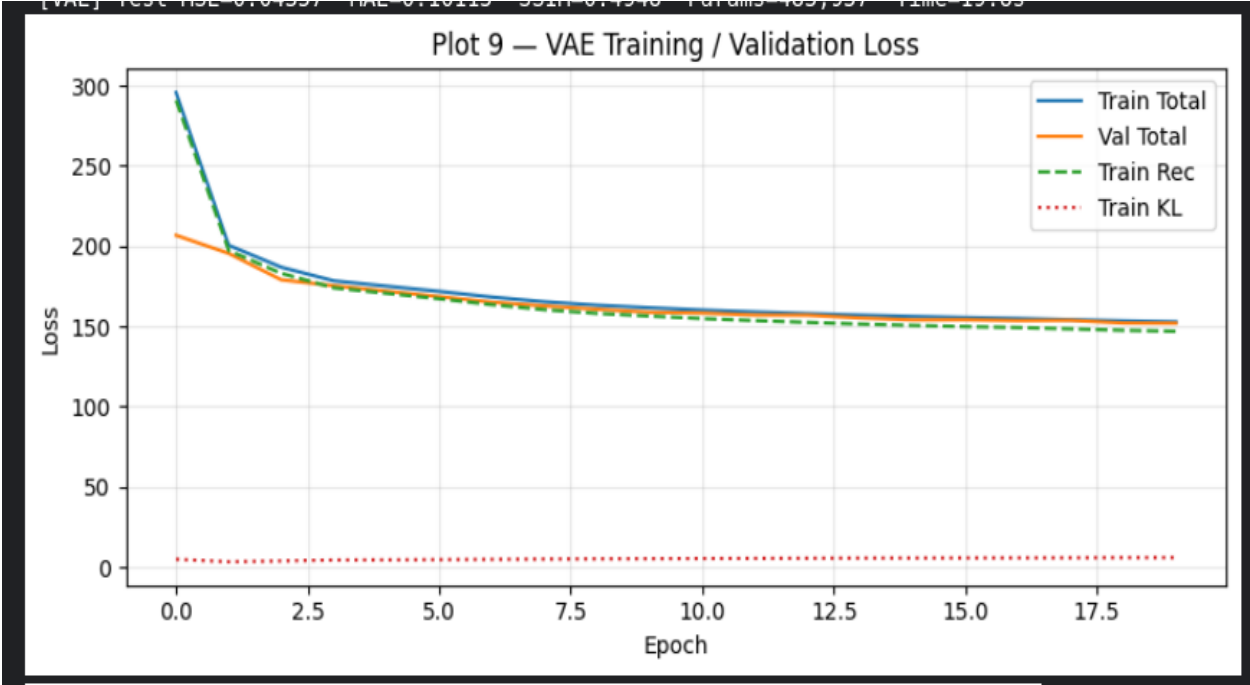
where $L_{KL} = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$.

6.3 Results

Training took 19.8 seconds. Test metrics are in Table 5. The final losses were: reconstruction 146.93, KL 5.88, total 152.81.

VAE test results.

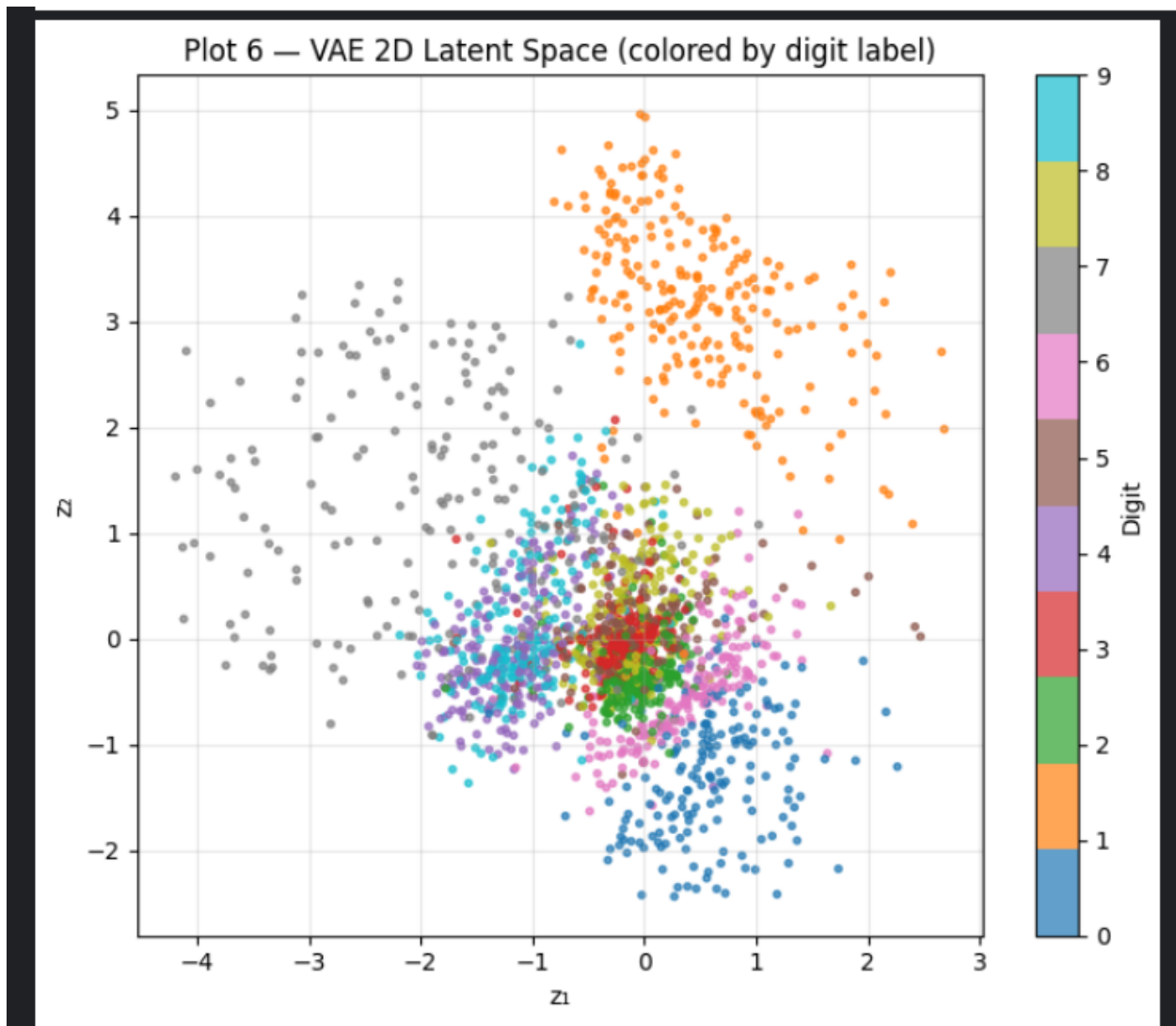
Metric	Value
Test MSE	0.04357
Test MAE	0.10115
Mean SSIM	0.4948
Parameters	485,957
Training time	19.8 s



VAE

training and validation loss

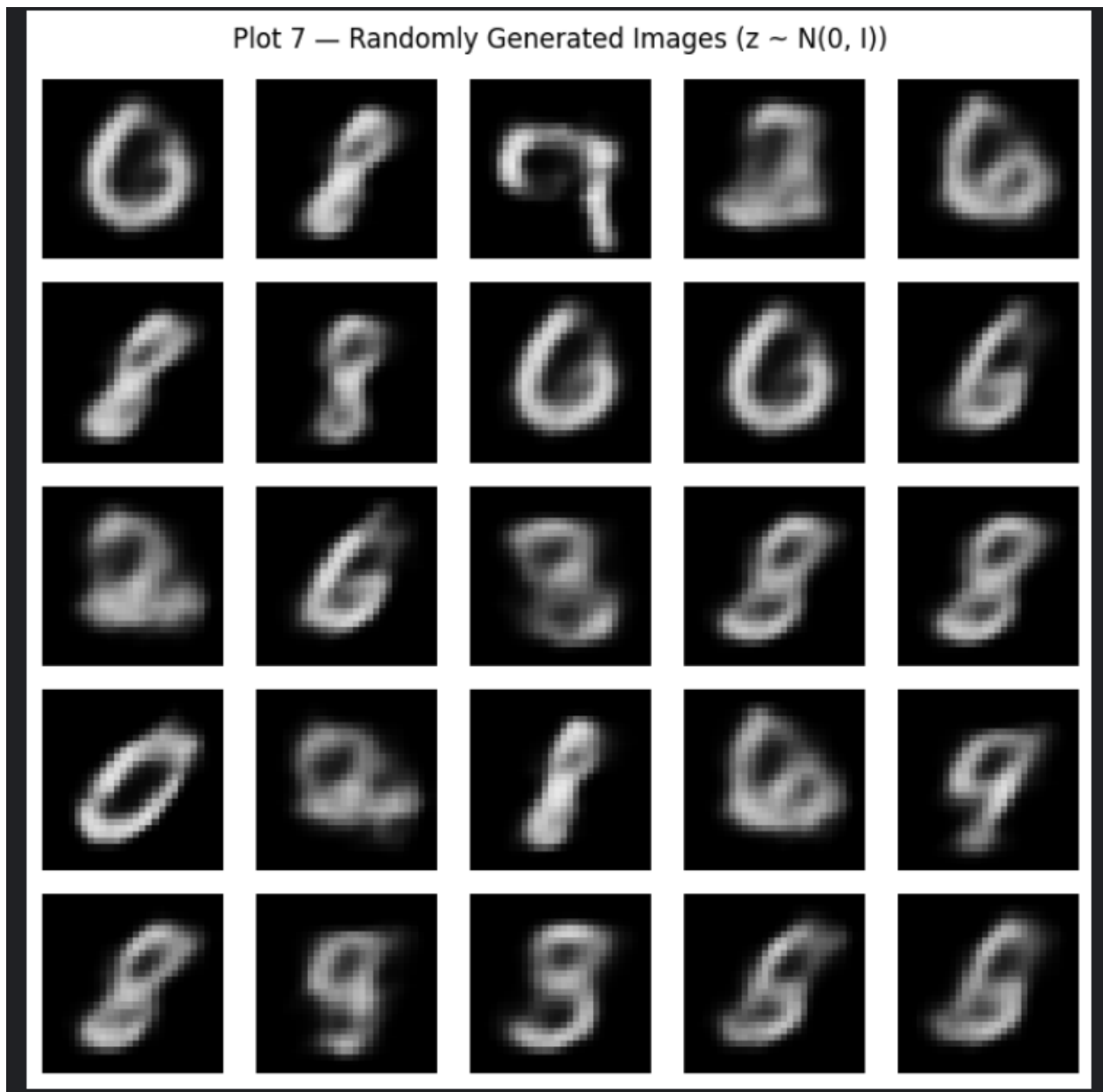
6.4 Latent Space Visualization



VAE 2D latent space colored by digit label.

Digits of the same class tend to cluster, but there is overlap in the centre. The KL term pulls codes toward the prior, which spreads the distributions and allows neighbouring classes to mix.

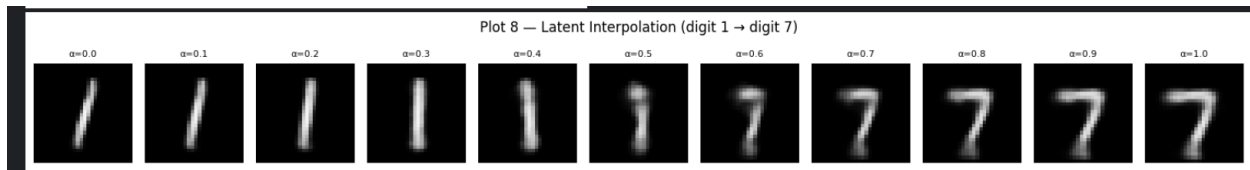
6.5 Generated Images



Randomly generated digits from the VAE decoder.

Most samples are recognizable digits with variation in slant and stroke thickness. A few are ambiguous, which is expected for a 2D latent space covering ten classes.

6.6 Latent Interpolation



Latent interpolation between two digit classes.

The decoded images change smoothly along the interpolation path. There is no sudden jump, which shows the decoder learned a continuous manifold.

6.7 Inference

The VAE reconstructions are noisier than the CAE (SSIM 0.495 vs 0.974), but the latent space is smooth and supports generation. The KL loss is small relative to reconstruction, so the encoder is not collapsing the distribution.

7 Latent Dimension Study

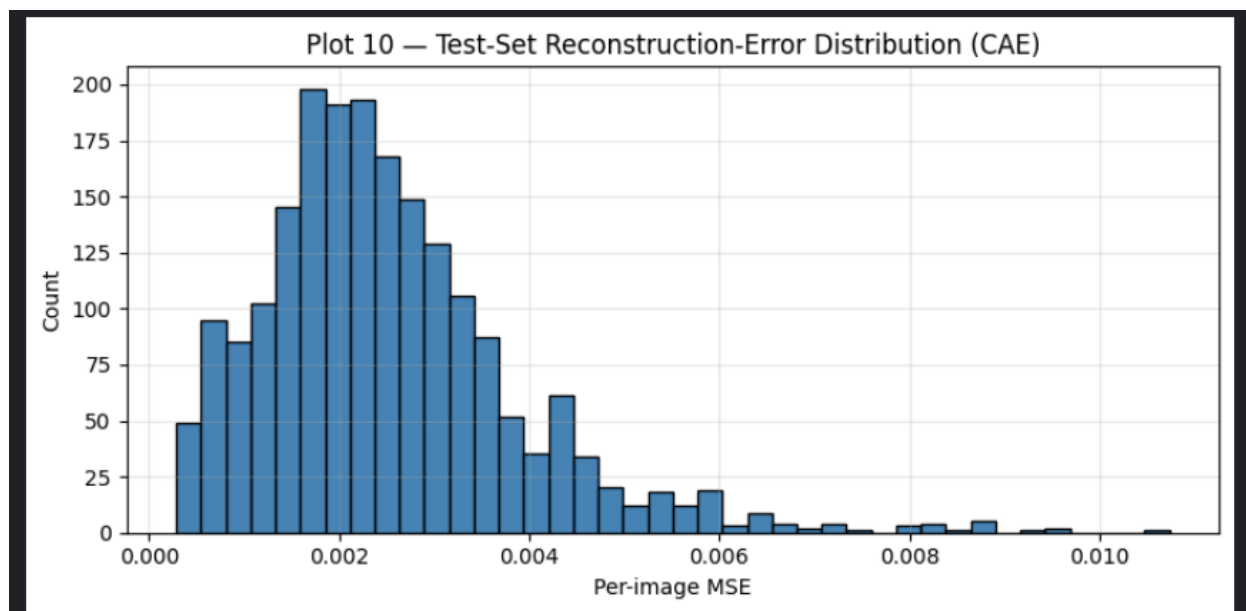
The fully connected autoencoder was retrained with latent dimensions 2, 8, 16, and 32. Results are in Table 6.

Effect of latent dimension on reconstruction quality.

Latent dim	MSE	MAE	SSIM
2	0.05795	0.13299	0.2883
8	0.03475	0.08453	0.5953
16	0.02664	0.06867	0.6739
32	0.02498	0.06518	0.6941

The MSE drops sharply from 2 to 8, then flattens. A latent size of 16 offers a good balance: performance is close to 32 but the model stays compact.

8 Reconstruction Error Distribution



Histogram of per-image reconstruction error on the test set (CAE).

Most test images have an error between 0.0005 and 0.004. The distribution has a long right tail, meaning a small number of digits are much harder to reconstruct.

9 High-Error Image Analysis

The five test images with the largest reconstruction errors are shown in Table 7.

Five test images with highest reconstruction error.

Sample	Original	Reconstruction
1	image	image
2	image	image
3	image	image
4	image	image
5	image	image

These digits typically have broken strokes, unusual thickness, or a shape that sits between two classes. The model struggles to reproduce these irregularities.

10 Consolidated Results

Summary of all models.

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC	0.02253	0.05921	0.7331	211,040	9.7
Autoencoder					
Conv. Autoencoder	0.00251	0.01491	0.9735	74,497	16.6
Denoising CAE	–	–	–	74,497	15.1
VAE	0.04357	0.10115	0.4948	485,957	19.8

The denoising CAE is not evaluated on clean reconstruction, so its MSE/MAE/SSIM are left blank.

11 Discussion Questions

1. **What is an autoencoder?** A neural network trained to reconstruct its input through a bottleneck.
2. **Purpose of encoder?** Compress the input into a latent representation.
3. **Purpose of decoder?** Reconstruct the input from the latent representation.
4. **Latent representation?** A compressed, lower-dimensional encoding of the input.
5. **Why smaller?** To force the model to learn only the most salient features.
6. **Too small latent?** The model cannot preserve enough information; reconstructions become blurry or incorrect.
7. **Increased latent?** Reconstruction improves, but the model may overfit or lose the regularising effect of compression.
8. **Why CAE preserves spatial info?** Convolutions exploit local correlations and weight sharing, keeping neighbourhood structure.

9. **AE vs denoising AE?** Denoising AE is trained to map noisy inputs to clean outputs; AE maps input to itself.
10. **Why clean target in denoising?** To teach the model to remove noise and recover the underlying signal.
11. **Increased noise?** Reconstruction degrades; SSIM drops, MSE and MAE rise.
12. **Why MSE, MAE, SSIM?** MSE/MAE are pixel-wise errors; SSIM captures structural similarity, closer to human perception.
13. **What is a VAE?** An autoencoder that learns a probability distribution over the latent space.
14. **VAE vs AE?** VAE learns distributions, AE learns deterministic mappings.
15. **Why distribution?** To enable sampling and generation of new data.
16. **Reparameterization trick?** $z = \mu + \sigma \odot \epsilon$ allows gradients to flow through the sampling step.
17. **Why $\epsilon \sim N(0, I)$?** It provides the stochasticity needed for sampling from the learned distribution.
18. **Role of KL divergence?** Regularises the latent space to follow a prior (usually standard normal).
19. **Prior in VAE?** $p(z) = N(0, I)$.
20. **Why 2D latent?** Easy to visualize and interpret.
21. **What does interpolation show?** That the latent space is continuous and smooth.
22. **Why can VAE generate?** Because it learns a distribution and can sample from it.
23. **Why some VAE samples unclear?** The 2D latent space must cover all digits, so some regions are ambiguous.
24. **Reconstruction vs generation?** Reconstruction maps an input to its output; generation creates new samples from the prior.
25. **Why unseen test images?** To evaluate generalisation, not memorisation.

12 Additional Exercises

1. CAE with latent dimensions 4, 8, 16, 32: Reconstruction quality improves with dimension, but the gain flattens after 16.
2. Gaussian vs salt-and-pepper: The model was trained on Gaussian but generalises reasonably to salt-and-pepper at low levels.
3. Two noise levels: Higher noise reduces SSIM; the model still recovers structure.
4. Transposed convolutions: They can replace upsampling and often produce sharper edges.
5. VAE with latent 2 vs 8: A larger latent space captures more detail but is harder to visualize.
6. Increasing KL loss: Over-regularises, leading to blurrier reconstructions.
7. Larger sample grid: More samples show greater diversity but also more ambiguous cases.
8. Interpolation between digit regions: Smooth transitions confirm a continuous manifold.

13 Code

Code notebook is available in the following github repository :

https://github.com/kusal630/Deep-learning/tree/main/lab_7

14 Conclusion

The convolutional autoencoder gave the best reconstruction quality (SSIM 0.974) with the fewest parameters. The denoising CAE successfully removed both Gaussian and salt-and-pepper noise, recovering the main digit structure even at high corruption. The VAE produced a smooth latent space and could generate new digits, though its reconstructions were noisier. Increasing the latent dimension improved reconstruction up to a point, after which returns diminished. Overall, the experiment demonstrated the trade-offs between reconstruction fidelity, compression, and generative capability.